Unified visual style for Figs. 1–5

All figures: 1000-unit canvas = 7.0 in (double column); Arial for text, Times italic for math; light fill + saturated stroke for modules.

Semantic palette

- **Perception / vision** — images, cameras, encoders
- **Slow system / VLM** — reasoning, planning
- **Fast system / action** — action expert, execution
- **Language / state tokens** — prompt, proprioception
- **Spiking / neuromorphic** — SNN, LIF neurons
- **Noise / neutral** — ε, inactive, background

event polarity ON / OFF

caution / key constraint (red, dashed)

Typography

(a) Panel title — 9 pt bold

Module label — 8 pt bold

Secondary text — 7 pt, grey

Annotation — 6.3 pt (minimum size)

Math: $x_\tau = (1\text{-}\tau)\varepsilon + \tau A$, $v_\theta(x_\tau, \tau \mid o)$, $a_{t:t+H}$

Convention used everywhere: τ = 0 noise → τ = 1 action

Components

VLM backbone
Gemma 2B

Action expert
300M, flow matching

Vision encoder
SigLIP

image tokens

text / state tokens

action tokens

noisy action tokens

data flow

interface (thick)

feedback / optional

planned motion

1 2 3 stage badges: perception · planning · execution

Illustration style (flat, side view)

